

SoilGrids in the future

When can SoilGrids at 100m resolution be expected?

Producing global soil information requires extensive infrastructure and resources. ISRIC has so far delivered complete and consistent global soil information products at 1km and 250m resolution. SoilGrids1km and SoilGrids250m are a step towards 100m global soil property maps. The GlobalSoilMap IUSS working group aims at delivering maps at a finer target resolution of 100m, and various countries have already done so. It is also on the agenda of the SoilGrids team.

Where can I find SoilGrids at a coarser resolution?

Aggregated SoilGrids products are available at a resolution of 1000 m and 5000 m. For this, the mean predictions from the 250 m products were aggregated at the corresponding coarser resolution. The data are available as COG (geotiff, [here](#)) from WebDAV (for instructions see [here](#)). We are working to produce modelling uncertainty at coarser resolution as well.

How is SoilGrids related to the GlobalSoilMap IUSS working group and its specifications?

SoilGrids aims to follow closely the [GlobalSoilMap specifications](#). It focuses exclusively on global predictions, currently with 250m being the finest resolution.

What happened to the map of soil types?

SoilGrids version 2.0 includes soil classification maps. These maps are not the result of new predictions, as with soil properties. They are an aggregation of the predictions of the previous release (2017, Hengl et al.).

SoilGrids version 2.0 serves only the Reference Soil Groups (RSG) of the World Reference Base for Soil Resources (WRB 2006), without differentiation by qualifier, in response to the feedback received from users. A single map is now created for each individual RSG. Each map shows the estimated probability of occurrence of the respective RSG within the map cell. This was obtained by summing the probabilities for each map predicted in 2017 that can be associated to the considered RSG for each map cell. Maps of probability occurrence for thirty different RSGs are available. Anthrosols, Technosols and Retisols are not considered, as they were not part of the predictions in SoilGrids version 2017.

An additional, 'summary' map provides for each cell the RSG's with highest probability of occurrence. It is called Most Probable and was obtained by selecting the RSG with the highest probability of occurrence for each map cell from the aggregated RSG probability maps described above. The RAT (Raster Attribute Table) with the legend of the classes is available in the VRT file and [here](#).

A new methodology is currently under development towards a unified soil classification map of the world, that would be more accurate and easier to use. However, the lack of reliable soil class observations remains an important issue. If you have access to such data, please consider contributing and help making a global soil classes map.

We are not working on new USDA soil Taxonomy classes as only a limited number of observations have the necessary USDA soil Taxonomy classification

Last updated on 2025-12-04